

About TVs

X17 (2017) — English translation

In a village with one street, a two-wire line is installed made of aluminum wire with a resistivity of $\rho = 2,8 \cdot 10^{-2} \text{ (}\Omega \cdot \text{mm}^2\text{)}/\text{m}$ and a cross-sectional area of $S = 4,0 \text{ cm}^2$. voltage at the input of line $U_0 = 220 \text{ V}$. Houses in the village stand every $l = 20 \text{ m}$; All houses have the same televisions with a nominal power $P_{\text{nom}} = 200 \text{ W}$ and voltage $U_{\text{nom}} = 220 \text{ V}$, each of which is connected to the network. The TV works normally if the voltage on it is not less than $U_{\text{min}} = 200 \text{ V}$.

A1 In how many houses N_0 can residents enjoy watching the news on TV at the same time? **3.50**
Assume that the resistance of the TV does not depend on the applied voltage, and the total number of houses in the village is $N \gg N_0$.

Source: <https://pho.rs/p/305>